Sarthak Roy

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PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience building production Generative AI applications - RAG pipelines, LLM-powered assistants, and vector-search systems using LangChain, Hugging Face, OpenRouter, and Pinecone. Complements this with strong production Computer Vision expertise: real-time detection, tracking, and multi-sensor fusion (YOLO, DINOv2, LoFTR, LiDAR-camera) deployed on live RTSP feeds via FastAPI. Skilled in Python-based ML pipeline development, API/backend integration, and cross-functional delivery of reliable, accuracy-focused AI systems for defense and enterprise applications.

SKILLS

AI/ML & GenAI: Python, Deep Learning, CNNs, OpenAI GPT-4, Hugging Face
RAG & LLM Systems: Retrieval-Augmented Generation (RAG), LangChain, Pinecone (vector database), OpenRouter, Embeddings, Context Retrieval
AI Automation & Workflows: n8n workflow automation, REST APIs, Webhooks, Real-time data pipelines
ML Engineering & APIs: FastAPI, Flask, REST API design, model inference pipelines, real-time inference
Cloud & Security: AWS deployments, OAuth, security protocols, threat monitoring
Computer Vision: OpenCV, YOLO, OC-SORT, DINOv2, LoFTR, object detection & tracking, 3D point cloud analysis, ROS2, LiDAR-camera sensor fusion, RTSP video pipelines
Backend / Full-Stack: React.js, Next.js, Node.js, Express.js, MongoDB, PostgreSQL, Firebase, Supabase, Nhost, WebSockets

EXPERIENCE

AI Engineer | Tardid Technologies Pvt. Ltd., Bangalore — *Feb 2026 - Present*

AI Engineer on Tardid's Brainbox AI product suite, building production ML/CV modules across three surveillance and situational-awareness platforms: Autonomous surveillance vessel, AI perimeter security, and Networked warfare management.

- **Real-time model inference & reliability:** Built production vessel detection and tracking (YOLO + OC-SORT) with motion-gating and per-track centroid history to suppress false positives, running continuously on a live RTSP video pipeline.
- **API & backend deployment:** Developed a vessel feature-matching and identification pipeline (LoFTR + DINOv2 descriptors, quality-adaptive thresholds, RANSAC filtering) and deployed it as a production service via a FastAPI + HTML frontend for operator review.
- **Accuracy & false-positive reduction:** Designed a classical CV intrusion/unattended-object detection pipeline (dual-EMA background subtraction, contour tracking) with diff-mask visualization and global lighting-change suppression to improve alert reliability on live CCTV feeds.
- **Systems integration:** Contributed to PTZ camera + LiDAR sensor fusion (ROS2) for multi-domain target tracking, resolving range/bearing accuracy issues to improve pipeline reliability.
- **Cross-functional delivery:** Collaborated with pipeline owners across detection, tracking, and zoom control to integrate individual ML/CV modules into production-ready, defense-grade systems.

Computer Vision Intern | Tardid Technologies Pvt. Ltd., Bangalore — *Aug 2023 - Oct 2023*

Navy defense project on real-time ship detection, tracking, and position estimation using deep learning and computer vision.

- **Model development:** Engineered CNN-based object detection and classification models for maritime environments, optimized for low visibility, fog, and occlusion.
- **Robustness under adverse conditions:** Designed image and video enhancement pipelines to improve visual clarity and detection accuracy in degraded conditions.
- **Applied research:** Researched and implemented SPAD (Single-Photon Avalanche Diode) and AOD (Acousto-Optic Deflection) principles for precision imaging and high-speed tracking.
- **Production integration:** Integrated AI-driven perception models with existing naval hardware and data pipelines, delivering a scalable, production-aligned CV solution with cross-functional teams.

PROJECTS

MedBot - AI Medical Knowledge Assistant (RAG)

Python (Flask, FastAPI), LangChain, Pinecone, OpenRouter, Hugging Face

- **Pipeline:** Built an end-to-end RAG pipeline for medical Q&A - processed medical PDFs into embeddings and indexed them in Pinecone for vector-based semantic retrieval.
- **Retrieval + generation:** Used LangChain to orchestrate context retrieval and pass relevant passages to LLMs (via OpenRouter/Hugging Face) for grounded response generation.
- **Delivery:** Exposed chatbot functionality through Flask/FastAPI backend APIs; improved Q&A clarity and reliability by grounding responses in retrieved medical document context.

SubSpace - AI-Powered Authentication & Chat Platform

React.js, Vite, TailwindCSS, Supabase/Nhost, n8n, OpenRouter, Hugging Face

- **AI chat integration:** Integrated OpenRouter API and Hugging Face API models to power an AI chat feature, achieving 85%+ response relevance.
- **Auth & backend:** Developed secure authentication with session handling/validation; integrated Supabase/Nhost backend with real-time workflow automation (n8n).
- **UI & impact:** Built a responsive glassmorphism UI (React.js, TailwindCSS); cut onboarding time by 60% and boosted engagement by 40%.

EDUCATION

B.Tech, Computer Science Engineering | Vellore Institute of Technology, Vellore | 2021 - 2025